

SUMIT SAH

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Education

Texas State University, Honors College

Expected: May 2027

B.Sc. Double Major: Computer Science & Applied Mathematics, Minor: Data Analytics | GPA: 3.85

San Marcos, TX

Relevant Coursework: Intro to Machine Learning, Probability & Statistics, Data Structures & Algorithms, Technical Writing

Licenses & Certifications: Stanford Online — Supervised Machine Learning; Google Data Analytics Professional Certificate

Honors & Awards: MAA Free Membership Nomination by Department, President's Honor Scholarship, Merry Kone FitzPatrick Endowed Scholarship, UG Honor for Academic Excellence, President's List (4x), Integration Bee Champion 2024

Experience

Undergrad Research Assistant — Machine Learning

Apr 2025 – Present

Department of Mathematics, Texas State University | Supervisor: Dr. Jeremiah Birrell

San Marcos, TX

- Designed and evaluated custom machine learning algorithms and deep neural networks (CNNs, ResNets) using PyTorch, applied to high-dimensional datasets.
- Collaborated with a team of 3 researchers to conduct systematic experiments, assess model architectures, and deliver weekly progress reports to guide performance tuning and validation.
- Investigating adversarial robustness through optimal transport-based metrics, quantifying model sensitivity to input perturbations and enhancing stability in both statistical and deep learning frameworks.

Data Analysis & Visualization Assistant

Sep 2025 – Present

University Libraries – Research & Data Services, Texas State University

San Marcos, TX

- Delivered 1:1 consultations and workshops on Python, R, and statistical methods, supporting students and faculty in applying data analysis techniques to academic research and coursework.
- Assisted in planning and organizing events like the Data & Donuts series, Datathon, and Open House, expanding participation and promoting data literacy.
- Maintained the Data Services web-page via LibGuide with new resources, increasing traffic and user engagement across campus.

Projects

Credit Risk Modeling (Loan Default Prediction) | github.com/sumit-sah314/credit-risk-modeling

- Engineered and trained classification models on the Taiwan Credit dataset (50K records, 33 features), improving default prediction accuracy by 12% over baseline.
- Applied Logistic Regression, Decision Trees, and XGBoost; optimized XGBoost achieved 97.1% accuracy and 0.98 AUC-ROC, demonstrating robustness on imbalanced data.
- Deployed an interactive Streamlit web app on Streamlit Cloud, enabling stakeholders to visualize predictions and compare model performance in real time.

RMF Bias Analysis (TXST Datathon 2025, 3rd Place) | github.com/sumit-sah314/professor-rating-bias-analysis

- Analyzed 20K+ RateMyProfessor reviews using Python (Pandas, Matplotlib) to uncover systematic biases in grading leniency, sampling, and review extremity.
- Generated 5+ visualizations highlighting key trends, enabling data-driven discussion around student evaluation reliability.
- Presented findings to a judging panel, earning 3rd place out of 6 teams by proposing actionable solutions to mitigate bias in academic evaluations.

Skills

Programming & Tools: Python, R, SQL, C/C++, Java, Git, Linux (HPC basics)

Machine Learning Frameworks : Scikit-learn, PyTorch, pandas, NumPy

Data Visualization: Matplotlib, Seaborn, Excel, MATLAB, Tableau, Power BI, MANIM(3b1b)

Deployment & Cloud: Streamlit Cloud, Docker, CI/CD Pipelines

Leadership & Extracurriculars

Math Club @ TXST – President: Lead 100+ members and organized 10+ academic events including AI talk, Calculus Bee, Escape Room, Workshops and Pi Day.

Content Creation – Produced 30+ YouTube videos with MANIM (1.1k+ subscribers, 50k+ views) on math, CS, and AI.

ACM AI – Member: Contributed in 6+ workshops, Kaggle contests, and 3 hackathons on AI ethics and neural networks.